

Linux Package Management - Complete Notes with Examples

I. Software Distribution Overview

Definition: Software packages are collections of files compiled into single units, managed by package managers.

Package Types

- **Debian packages (.deb):** Used by Debian, Ubuntu, Linux Mint, etc.
- **Red Hat packages (.rpm):** Used by RHEL, Fedora, CentOS, etc.

Package Ecosystem

- **Upstream providers:** Original software developers who create and compile the code
- **Package maintainers:** Review, manage, and distribute software as packages
- **End users:** Install packages through package managers

Example Package Names:

```
# Debian package example
google-chrome-stable_current_amd64.deb

# RPM package example
firefox-91.0-1.fc34.x86_64.rpm
```

2. Package Repositories

Definition: Central storage locations for packages, typically hosted on the internet.

Repository Structure Example

```
WackyWidgets Repository: http://download.widgets/linux/deb/
├─ CoolWidget.deb
├─ SuperWidget.deb
```

```
└─ UltraWidget.deb
└─ MegaWidget.deb
```

Repository Configuration

Debian systems - Sources file location:

```
$ cat /etc/apt/sources.list
# Main Ubuntu repositories
deb http://archive.ubuntu.com/ubuntu/ focal main restricted
deb http://archive.ubuntu.com/ubuntu/ focal-updates main restricted
deb http://archive.ubuntu.com/ubuntu/ focal universe
deb http://archive.ubuntu.com/ubuntu/ focal multiverse

# Security updates
deb http://security.ubuntu.com/ubuntu/ focal-security main restricted
```

Adding custom repositories:

```
# Add a new repository source
$ echo "deb http://download.widgets/linux/deb/ /" | sudo tee -a
/etc/apt/sources.list

# Update package list after adding repository
$ sudo apt update
```

3. File Archiving and Compression

gzip Compression

Compress a file:

```
$ gzip mycoolfiler
# Creates: mycoolfiler.gz (original file is replaced)

$ ls
mycoolfiler.gz
```

Decompress a file:

```
$ gunzip mycoolfiler.gz
# Restores: mycoolfiler (compressed file is removed)
```

```
$ ls
mycoolfiler
```

Keep original file while compressing:

```
$ gzip -k mycoolfiler
# Creates: mycoolfiler.gz (keeps original mycoolfiler)

$ ls
mycoolfiler  mycoolfiler.gz
```

tar Archiving

Create archive:

```
$ tar cvf mytarfile.tar mycoolfiler1 mycoolfiler2 mycoolfiler3
# Creates: mytarfile.tar containing all specified files

$ ls
mytarfile.tar  mycoolfiler1  mycoolfiler2  mycoolfiler3
```

Archive flags explanation:

- **c**: Create archive
- **v**: Verbose output (show progress)
- **f**: Specify filename (must be last flag)

Extract archive:

```
$ tar xvf mytarfile.tar
# Extracts all files from mytarfile.tar

$ ls
mytarfile.tar  mycoolfiler1  mycoolfiler2  mycoolfiler3
```

Extract flags explanation:

- **x**: Extract files
- **v**: Verbose output
- **f**: Specify filename

Combined tar and gzip Operations

Create compressed archive:

```
$ tar czf myarchive.tar.gz file1 file2 file3
# Creates compressed archive in one step

$ ls
myarchive.tar.gz  file1  file2  file3
```

Extract compressed archive:

```
$ tar xzf myarchive.tar.gz
# Extracts and decompresses in one step

$ ls
myarchive.tar.gz  file1  file2  file3
```

Additional tar flags:

- **z**: Use gzip compression/decompression
- **j**: Use bzip2 compression/decompression

Memory aid: "eXtract all Zee Files!" (xzf)

Other Archive Types

Common formats you might encounter:

```
# bzip2 compression
$ bzip2 filename
$ bunzip2 filename.bz2

# ZIP archives
$ zip archive.zip file1 file2 file3
$ unzip archive.zip

# View archive contents without extracting
$ tar tzf archive.tar.gz      # List contents of compressed tar
$ tar tvf archive.tar        # List contents of tar archive
```

4. Package Dependencies

Definition: Additional packages or libraries required for a package to function properly.

Dependency Relationship Example

```
Main Package: LibreOffice
├─ Dependency: Java Runtime Environment (JRE)
├─ Dependency: GTK+ Libraries
├─ Dependency: FontConfig
└─ Dependency: Various system libraries
```

Types of Dependencies

- **Package dependencies:** Other software packages
- **Library dependencies:** Shared code libraries
- **System dependencies:** Core system components

Example dependency chain:

```
# Example: Installing a media player might require:
vlc → libvlc → ffmpeg → libx264 → various codecs
```

Dependency Problems

```
# Without proper dependency management:
$ dpkg -i some-package.deb
dpkg: dependency problems prevent configuration of some-package:
some-package depends on libexample1 (>= 1.2.3); however:
Package libexample1 is not installed.
```

5. Low-level Package Management - rpm and dpkg

Definition: Basic package management tools that handle individual package files but don't resolve dependencies automatically.

dpkg (Debian Package Manager)

Install package:

```
$ sudo dpkg -i google-chrome-stable_current_amd64.deb
Selecting previously unselected package google-chrome-stable.
(Reading database ... 185432 files and directories currently installed.)
```

```
Preparing to unpack google-chrome-stable_current_amd64.deb ...
Unpacking google-chrome-stable (91.0.4472.164-1) ...
Setting up google-chrome-stable (91.0.4472.164-1) ...
```

Remove package:

```
$ sudo dpkg -r google-chrome-stable
(Reading database ... 185440 files and directories currently installed.)
Removing google-chrome-stable (91.0.4472.164-1) ...
```

List installed packages:

```
$ dpkg -l
Desired=Unknown/Install/Remove/Purge/Hold
| Status=Not/Inst/Conf-files/Unpacked/halF-conf/Half-inst/trig-aWait/Trig-
pend
|/ Err?=(none)/Reinst-required (Status,Err: uppercase=bad)
||/ Name          Version          Architecture Description
+++-----+-----+-----+
=====
ii  bash             5.0-6ubuntu1     amd64          GNU Bourne Again SHell
ii  coreutils        8.30-3ubuntu2    amd64          GNU core utilities
```

Query specific package:

```
$ dpkg -l | grep chrome
ii google-chrome-stable 91.0.4472.164-1 amd64 The web browser from
Google
```

Get package information:

```
$ dpkg -s google-chrome-stable
Package: google-chrome-stable
Status: install ok installed
Priority: optional
Section: web
Installed-Size: 272748
Maintainer: Chrome Linux Team <chromium-dev@chromium.org>
Architecture: amd64
Version: 91.0.4472.164-1
```

rpm (Red Hat Package Manager)

Install package:

```
$ sudo rpm -i firefox-91.0-1.fc34.x86_64.rpm
Preparing... #####
[100%]
Updating / installing...
 1:firefox-91.0-1.fc34 #####
[100%]
```

Remove package:

```
sudo rpm -e firefox
```

List all installed packages:

```
$ rpm -qa
firefox-91.0-1.fc34.x86_64
bash-5.1.8-2.fc34.x86_64
coreutils-8.32-30.fc34.x86_64
```

Query specific package:

```
$ rpm -qa | grep firefox
firefox-91.0-1.fc34.x86_64
```

Get package information:

```
$ rpm -qi firefox
Name           : firefox
Version        : 91.0
Release        : 1.fc34
Architecture    : x86_64
Install Date    : Wed 28 Jul 2021 10:30:45 AM EDT
Group           : Applications/Internet
Size            : 235891567
License        : MPLv1.1 or GPLv2+ or LGPLv2+
Signature       : RSA/SHA256, Mon 12 Jul 2021 01:23:45 PM EDT
Source RPM      : firefox-91.0-1.fc34.src.rpm
Build Date      : Mon 12 Jul 2021 12:45:23 PM EDT
Build Host      : buildvm-x86-01.iad2.fedoraproject.org
Packager        : Fedora Project
Vendor          : Fedora Project
URL             : https://www.mozilla.org/firefox/
```

Summary : Mozilla Firefox Web browser
Description : Mozilla Firefox is an open-source web browser, designed for standards compliance, performance and portability.

6. High-level Package Management - yum and apt

Definition: Advanced package management systems that handle dependencies automatically.

apt (Advanced Package Tool) - Debian Family

Update package repository information:

```
$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu focal InRelease
Get:2 http://archive.ubuntu.com/ubuntu focal-updates InRelease [114 kB]
Get:3 http://archive.ubuntu.com/ubuntu focal-backports InRelease [101 kB]
Get:4 http://security.ubuntu.com/ubuntu focal-security InRelease [114 kB]
Fetched 329 kB in 2s (164 kB/s)
Reading package lists... Done
```

Install package:

```
$ sudo apt install firefox
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  firefox-locale-en libxul0d
The following NEW packages will be installed:
  firefox firefox-locale-en libxul0d
0 upgraded, 3 newly installed, 0 to remove and 0 not upgraded.
Need to get 54.7 MB of archives.
After this operation, 196 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

Remove package:

```
$ sudo apt remove firefox
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages will be REMOVED:
  firefox firefox-locale-en
```

```
0 upgraded, 0 newly installed, 2 to remove and 0 not upgraded.  
After this operation, 196 MB disk space will be freed.  
Do you want to continue? [Y/n] y
```

Upgrade all packages:

```
$ sudo apt upgrade  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
Calculating upgrade... Done  
The following packages will be upgraded:  
  libc6 libc6-dev libc-dev-bin  
3 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.  
Need to get 9,463 kB of archives.  
After this operation, 0 B of additional disk space will be used.  
Do you want to continue? [Y/n] y
```

Search for packages:

```
$ apt search text-editor  
Sorting... Done  
Full Text Search... Done  
gedit/focal 3.36.2-1 amd64  
  official text editor of the GNOME desktop environment  
  
vim/focal 2:8.1.2269-1ubuntu5 amd64  
  Vi IMproved - enhanced vi editor
```

Show package information:

```
$ apt show firefox  
Package: firefox  
Version: 91.0+build2-0ubuntu1  
Priority: optional  
Section: web  
Origin: Ubuntu  
Maintainer: Ubuntu Mozilla Team <ubuntu-mozillateam@lists.ubuntu.com>  
Bugs: https://bugs.launchpad.net/ubuntu/+filebug  
Installed-Size: 196 MB  
Depends: libc6 (>= 2.17), libxul0d (= 91.0+build2-0ubuntu1)  
Homepage: https://www.mozilla.org/firefox/  
Description: Mozilla Firefox web browser
```

yum (Yellowdog Updater Modified) - Red Hat Family

Update package repository and upgrade packages:

```
$ sudo yum update
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
* base: mirror.example.com
* extras: mirror.example.com
* updates: mirror.example.com
Resolving Dependencies
--> Running transaction check
---> Package firefox.x86_64 0:68.12.0-1.el7.centos will be updated
---> Package firefox.x86_64 0:91.0-1.el7.centos will be an update
--> Finished Dependency Resolution
```

Install package:

```
$ sudo yum install firefox
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
Resolving Dependencies
--> Running transaction check
---> Package firefox.x86_64 0:91.0-1.el7.centos will be installed
--> Processing Dependency: mozilla-filesystem for package: firefox-91.0-1.el7.centos.x86_64
--> Running transaction check
---> Package mozilla-filesystem.x86_64 0:1.9-11.el7 will be installed
--> Finished Dependency Resolution
```

Remove package:

```
$ sudo yum erase firefox
Loaded plugins: fastestmirror, langpacks
Resolving Dependencies
--> Running transaction check
---> Package firefox.x86_64 0:91.0-1.el7.centos will be erased
--> Finished Dependency Resolution
```

Search for packages:

```
$ yum search text-editor
Loaded plugins: fastestmirror, langpacks
```

```
Loading mirror speeds from cached hostfile
===== N/S matched: text-editor
=====
gedit.x86_64 : A text editor for GNOME
vim-enhanced.x86_64 : A version of the VIM editor which includes recent
enhancements
```

Get package information:

```
$ yum info firefox
Loaded plugins: fastestmirror, langpacks
Loading mirror speeds from cached hostfile
Available Packages
Name           : firefox
Arch           : x86_64
Version        : 91.0
Release        : 1.el7.centos
Size           : 102 M
Repo           : updates/7/x86_64
Summary        : Mozilla Firefox Web browser
URL            : https://www.mozilla.org/firefox/
License        : MPLv1.1 or GPLv2+ or LGPLv2+
Description    : Mozilla Firefox is an open-source web browser, designed for
                  : standards compliance, performance and portability.
```

7. Compiling from Source Code

Definition: Building software directly from source code when pre-compiled packages aren't available.

Prerequisites

Install build tools:

```
# Debian/Ubuntu
$ sudo apt install build-essential
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
  build-essential dpkg-dev fakeroot g++ g++-9 gcc gcc-9 libc6-dev libdpkg-
perl make
0 upgraded, 10 newly installed, 0 to remove and 0 not upgraded.
```

```
# Red Hat/CentOS/Fedora
$ sudo yum groupinstall "Development Tools"
```

Basic Compilation Process

Step 1: Extract source code

```
$ tar -xzf awesome-software-1.2.3.tar.gz
awesome-software-1.2.3/
awesome-software-1.2.3/README
awesome-software-1.2.3/INSTALL
awesome-software-1.2.3/configure
awesome-software-1.2.3/Makefile.in
awesome-software-1.2.3/src/
awesome-software-1.2.3/src/main.c

$ cd awesome-software-1.2.3
```

Step 2: Read documentation

```
$ cat README
AWESOME SOFTWARE v1.2.3
=====

This software requires:
- GCC compiler
- libawesome-dev (>= 2.1)
- libcool-dev (>= 1.5)

To compile and install:
1. ./configure
2. make
3. make install

$ cat INSTALL
# Additional installation instructions...
```

Step 3: Configure build

```
$ ./configure
checking for gcc... gcc
checking whether the C compiler works... yes
checking for library dependencies... yes
```

```
checking for libawesome... yes
checking for libcool... yes
configure: creating ./config.status
config.status: creating Makefile
config.status: creating config.h
```

Handle missing dependencies:

```
$ ./configure
checking for gcc... gcc
checking for libawesome... no
configure: error: libawesome development package not found

# Install missing dependency
$ sudo apt install libawesome-dev
$ ./configure
# Now should succeed
```

Step 4: Compile the software

```
$ make
gcc -c -o main.o main.c
gcc -c -o utils.o utils.c
gcc -o awesome-software main.o utils.o -lawesome -lcool
```

Step 5: Install the software

```
$ sudo make install
install -d /usr/local/bin
install -m 755 awesome-software /usr/local/bin/
install -d /usr/local/share/man/man1
install -m 644 awesome-software.1 /usr/local/share/man/man1/
```

Alternative: Using checkinstall

Install checkinstall:

```
sudo apt install checkinstall
```

Use checkinstall instead of make install:

```
$ sudo checkinstall
checkinstall 1.6.2, Copyright 2009 Felipe Eduardo Sanchez Diaz Duran
This software is released under the GNU GPL.
```

```
*****
**** Debian package creation selected ****
*****
```

This package will be built according to these values:

```
0 - Maintainer: [ user@hostname ]
1 - Summary: [ Package created with checkinstall 1.6.2 ]
2 - Name: [ awesome-software ]
3 - Version: [ 1.2.3 ]
4 - Release: [ 1 ]
5 - License: [ GPL ]
6 - Group: [ checkinstall ]
7 - Architecture: [ amd64 ]
8 - Source location: [ awesome-software-1.2.3 ]
9 - Alternate source location: [ ]
10 - Requires: [ ]
11 - Provides: [ awesome-software ]
12 - Conflicts: [ ]
13 - Replaces: [ ]
```

Enter a number to change any of them or press ENTER to continue:

Installing with `make install`...

===== Installation successful =====

Copying documentation directory...

./

./README

./INSTALL

Copying files to the temporary directory...Done.

Stripping ELF binaries and libraries...Done.

Compressing `man` pages...Done.

Building `file` list...Done.

Building Debian package...Done.

Installing Debian package...Done.

```
Erasing temporary files...Done.
```

```
Writing backup package...Done.
```

```
Deleting temp dir...Done.
```

```
*****
```

```
Done. The new package has been installed and saved to
```

```
/home/user/awesome-software_1.2.3-1_amd64.deb
```

```
You can remove it from your system anytime using:
```

```
dpkg -r awesome-software
```

```
*****
```

Uninstallation

With **regular make install**:

```
$ sudo make uninstall
rm -f /usr/local/bin/awesome-software
rm -f /usr/local/share/man/man1/awesome-software.1
```

With **checkinstall (recommended)**:

```
$ sudo dpkg -r awesome-software
(Reading database ... 185440 files and directories currently installed.)
Removing awesome-software (1.2.3-1) ...
```

Summary Commands Quick Reference

Package Installation

# Debian/Ubuntu	
sudo apt update	# Update package lists
sudo apt install package_name	# Install package
sudo apt remove package_name	# Remove package
sudo apt upgrade	# Upgrade all packages
apt search keyword	# Search for packages

```
apt show package_name           # Show package info

# Red Hat/CentOS/Fedora
sudo yum update                 # Update packages
sudo yum install package_name   # Install package
sudo yum erase package_name     # Remove package
yum search keyword              # Search for packages
yum info package_name           # Show package info
```

Direct Package Installation

```
# Debian packages
sudo dpkg -i package.deb        # Install .deb package
sudo dpkg -r package_name       # Remove package
dpkg -l                         # List installed packages
dpkg -s package_name            # Show package status

# RPM packages
sudo rpm -i package.rpm         # Install .rpm package
sudo rpm -e package_name        # Remove package
rpm -qa                         # List all packages
rpm -qi package_name            # Show package info
```

Archive Operations

```
# tar operations
tar cvf archive.tar files...    # Create archive
tar xvf archive.tar             # Extract archive
tar czf archive.tar.gz files... # Create compressed archive
tar xzf archive.tar.gz          # Extract compressed archive

# gzip operations
gzip filename                   # Compress file
gunzip filename.gz              # Decompress file
```

Source Compilation

```
# Standard compilation
sudo apt install build-essential # Install build tools
tar -xzf package.tar.gz          # Extract source
cd package-directory              # Enter directory
./configure                      # Configure build
make                             # Compile
```

```
sudo make install          # Install

# With checkinstall (recommended)
sudo checkinstall          # Install and create package
sudo dpkg -r package_name  # Remove later
```